The Ising perceptron at nonzero margin: certified bounds along the replica-symmetric curve

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Abstract

The storage capacity of the Ising perceptron at margin κ is predicted by a replica-symmetric variational problem. Huang proved that the replica-symmetric threshold $\alpha_\star(\kappa)$ is an upper bound at every margin where four low-dimensional conditions hold. We verify the four conditions in ball arithmetic at $\kappa = -0.45$, -0.05, 0.05, 0.0995 and 0.13, so the bound holds there unconditionally; at $\kappa = 0.05$ the capacity is at most 0.781075. Three of the conditions hold on the strip $\kappa \in [-0.65, 0.19]$. Away from zero margin the threshold is known only within an interval, and over that interval the fixed point of the replica-symmetric equations is enclosed only in a box about $8 \cdot 10^3$ times wider in one coordinate than at $\kappa = 0$; at that width the zero-margin proof of one of the conditions fails. We write the differences of the log-partition function that this condition needs against a point at which they vanish identically, so that their enclosures shrink in proportion to the distance from that point. Whether $\alpha_\star(\kappa)$ is also a lower bound at $\kappa \neq 0$ is open.

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1 Introduction

Store $m = \lceil \alpha N \rceil$ random patterns in a linear classifier with binary weights: for i.i.d. standard Gaussian vectors $g_1, \dots, g_m$ in $\mathbb{R}^N$ and a margin $\kappa \in \mathbb{R}$, ask whether some $w \in \{-1, +1\}^N$ satisfies

$$\frac{\langle g_a, w \rangle}{\sqrt{N}} \;\geq\; \kappa \qquad \text{for all } a \leq m.$$

We call this the margin condition. Write $m_N(\kappa)$ for the largest m for which such a w exists. The capacity at margin κ is at most α if for every $\alpha' > \alpha$, $m_N(\kappa) < \alpha' N$ with probability $1 - o(1)$ as $N \to \infty$. Krauth and Mézard [1] computed the replica-symmetric prediction $\alpha_\star(\kappa)$ for the capacity and conjectured it sharp; at $\kappa = 0$ their constant is $\alpha_\star = 0.8330786\ldots$. Rigorous bounds at zero margin go back to Kim and Roche [4] and Talagrand [5], who proved that the capacity is at most $1 - \varepsilon$ for some $\varepsilon > 0$; Altschuler and Tikhomirov [12] give a complete proof of the bound 0.847 outlined by Krauth and Mézard. At zero margin Ding and Sun [9] proved the lower bound with positive probability, which the sharp-threshold results of Xu [11]

and of Nakajima and Sun [20] raise to probability $1 - o(1)$, and Huang [10] proved the upper bound, each under numerical conditions on explicit low-dimensional variational problems; these conditions are verified in ball arithmetic in [14].

Huang's upper-bound theorem [10, Theorem 3.6], however, is stated at every margin. For each κ it bounds the capacity by $\alpha_\star(\kappa)$, the root of the replica-symmetric free energy along its fixed-point branch, conditionally on four numerical conditions at that margin. The first asks that the fixed-point equations have a unique solution, along which the free energy changes sign. The second asks that the associated approximate-message-passing iteration be stable at that solution. The third asks that an explicit function of one real variable take a negative value at some point. The fourth asks that a function $\mathcal{S}_\star$ of two real variables be nonpositive on the whole plane (Conditions 1–4 of Section 2). Stojnic [21] writes the capacity at margin κ as the value of a lifted random-duality functional; assuming the stationarized fully lifted frame of his theory at the second level of lifting [21, Theorem 3], he obtains the threshold $\alpha_\star(\kappa)$ at every κ, and reports numerically that the third level of lifting gives the same value. The Main Theorem is unconditional: it is Huang's theorem with its four conditions proved. We verify all four conditions at five nonzero margins.

Main Theorem (Theorem 4)**.** *For $\kappa \in \{-0.45,\ -0.05,\ 0.05,\ 0.0995,\ 0.13\}$ the four conditions of Huang's theorem hold, and the threshold $\alpha_\star(\kappa)$ lies in the interval*

$$\begin{aligned} &\kappa = -0.45: \ [1.557652103,\ 1.557654133], \quad \kappa = -0.05: \ [0.889408031,\ 0.889409783],\\ &\kappa = 0.05: \ [0.781073068,\ 0.781074776], \quad \kappa = 0.0995: \ [0.733478513,\ 0.733480205],\\ &\kappa = 0.13: \ [0.705932217,\ 0.705933898]. \end{aligned}$$

In particular the capacity at margin κ is at most $\alpha_\star(\kappa)$, hence at most the right end of the interval.

Let (q_0, ψ_0) be the fixed point of the replica-symmetric equations at $\alpha = \alpha_\star(\kappa)$ (Section 2), let $\dot{H} \sim N(0, \psi_0)$ and $M = \tanh \dot{H}$. The fourth condition asks that $\mathcal{S}_\star(\lambda) \le 0$ for every $\lambda \in \mathbb{R}^2$, where $\mathcal{S}_\star(\lambda)$ is an infimum over $s \ge 0$ of a functional of $\Lambda = \tanh(\lambda_1 \dot{H} + \lambda_2 M)$ and s; equality holds at $\lambda = (1, 0)$, where $\Lambda = M$. At $\kappa = 0$ it is proved in [14] through the stronger inequality over all functions Λ of $\dot{H}$ with values in $[-1, 1]$, by bounding the entropy of Λ through the two moments $a = (\mathbb{E}[\dot{H}\Lambda], \mathbb{E}[M\Lambda])$ and the log-partition function

$$\Phi(b) = \mathbb{E} \log 2\cosh(b_1 \dot{H} + b_2 M), \qquad b \in \mathbb{R}^2 :$$

the entropy is at most $\Phi(b) - b \cdot a$ for every b. Near the moments $a^\star = (\psi_0(1 - q_0), q_0)$ of M the proof has to locate the optimal b, and for this it needs the signs of differences $\Phi(\lambda) - \Phi(\lambda') - (\lambda - \lambda') \cdot a$, with $\lambda, \lambda' \in \mathbb{R}^2$, that are of order 10^{-7}. At $\kappa = 0$ the fixed point is enclosed to width $8 \cdot 10^{-10}$ in q_0 and $1.2 \cdot 10^{-8}$ in ψ_0, and separate enclosures of the values of Φ decide these signs. At $\kappa \ne 0$ the threshold is known only within an interval, of width $1.7 \cdot 10^{-6}$ at $\kappa = 0.05$, found by a certified bisection on the sign of the function whose zero defines the threshold and limited by the width of its enclosure near that zero (Section 3.4), and the conditions must hold at every α of that interval. The fixed point then ranges over a box of width $1.5 \cdot 10^{-6}$ in q_0 and $9.2 \cdot 10^{-5}$ in ψ_0: it moves by $2.4 \cdot 10^{-5}$ in ψ_0 across the interval, and at each end ψ_0 is a Gaussian integral, enclosed by a Riemann sum of 4000 cells to width $6.9 \cdot 10^{-5}$. Over this box an enclosure of a value of Φ has width about $2 \cdot 10^{-5}$, two hundred times the differences whose signs are needed, and the proof of [14] does not apply.

We take the differences against the point $b_0 = (1, 0)$, at which $\tanh(b_1\dot{H} + b_2 M) = M$. At a fixed point $\mathbb{E}[M^2] = q_0$, and Gaussian integration by parts gives $\nabla\Phi(b_0) = (\mathbb{E}[\dot{H}M], \mathbb{E}[M^2]) = a^\star$. For every $\lambda \in \mathbb{R}^2$,

$$\Phi(\lambda) - \Phi(b_0) - (\lambda - b_0)\cdot\nabla\Phi(b_0) = \mathbb{E}\big[\log\big(\cosh\Delta + M\sinh\Delta\big) - M\Delta\big], \qquad \Delta = (\lambda_1 - 1)\dot{H} + \lambda_2 M. \tag{1}$$

The right side vanishes at $\lambda = b_0$ for every ψ_0 and is of second order in $|\lambda - b_0|$; its enclosure over the box, an interval evaluation of the integrand, has width proportional to $|\lambda - b_0|$: at distance 0.01 from b_0 it is 57 to 251 times narrower, depending on the direction, than a difference of two enclosures of values of Φ. With $a^\star$ carried as the function $(\psi_0(1 - q_0), q_0)$ on the box, the signs are decided. Two further estimates are needed. Near $a^\star$ the inequality is proved along the segments $a^\star + tv$, $t \geq 0$, through bounds on second derivatives in t. These involve the inverse Mills ratio E of the standard Gaussian and its first two derivatives at arguments V as large as 20, where $E(V) - V$ is a difference of two large enclosures; the bounds $0 < E(V) - V < 1/V$, $0 < E' < 1$ and $0 < E'' < E(V) - V$ (Lemma 6) take its place. And the fourth condition carries a second variable $s \geq 0$, over which an infimum is taken, with optimal value $s_0 = \sqrt{1 - q_0}$ at $a^\star$; along each segment we take $s = s_0 + ct$ with a slope c fixed by the direction v alone, so that the second-derivative bounds on successive intervals of t chain from $t = 0$ (Section 4.2). With these changes the proof of [14] applies at $\kappa = -0.45$, -0.05, 0.05, 0.0995 and 0.13.

Three things are different at $\kappa \neq 0$ besides the width. Of the two relations behind Huang's closed forms at zero margin, one fails at $\kappa \neq 0$ (Section 4.4). On the whole strip $\kappa \in [-0.65, 0.19]$ the fixed point is unique in the rectangle of Condition 1, but the bound on the derivative of the fixed-point map that gives uniqueness for $\kappa \in [-0.65, 0.10]$ exceeds 1 near $\kappa = 0.12$; on $(0.10, 0.19]$ uniqueness follows from Nakajima's theorem [13, Theorem 1], whose premise we verify in ball arithmetic, and from a contraction bound on narrower rectangles about the fixed point (Section 3.4). And the first three conditions hold on intervals of κ (Theorem 3), while the fourth is proved at points: over a slab, an interval of κ of width $5 \cdot 10^{-4}$, the fixed point ranges over about $3 \cdot 10^{-3}$ in q_0, two thousand times its range at a point (Section 4.3.1).

Theorem 3 covers the strip by its 1680 slabs. Its proof uses the formula for $\partial R/\partial q$ of Lemma 2, which is Nakajima's formula [13, eq. (18)] rewritten for R, and the partial derivatives of the replica-symmetric free energy of Lemma 1, which extend the computation of Ding and Sun [9, Lemma 7.1 and Corollary 7.2] off the fixed-point branch. For $\kappa < 0$, Section 5 gives the lower bound $\max\{\alpha_\star(0), \alpha_c^{\mathrm{sym}}(|\kappa|)\}$, where $\alpha_c^{\mathrm{sym}}(r)$ is the threshold of the symmetric binary perceptron at width r, by monotonicity and by embedding a symmetric perceptron; it is below $\alpha_\star(\kappa)$, by 0.056 at $\kappa = -0.05$ and by 0.72 at $\kappa = -0.45$.

Section 2 states the replica-symmetric system, Huang's four conditions and two identities. Section 3 proves Conditions 1–3 on the strip. Section 4 proves the Main Theorem: the function of Condition 4 at general margin, the certificates at the five margins, the Hessian at the maximizer, why intervals of κ are out of reach (Section 4.3.1), and which inequalities are restated over the integers (Section 4.5). Section 5 shows that the second moment of the number of solutions is exponentially larger than the square of the first at every margin and gives the lower bound for $\kappa < 0$. The notation is collected at the end.

2 The system and the four conditions

Let φ and Ψ denote the standard Gaussian density and upper tail, $E = \varphi/\Psi$ the inverse Mills ratio (an increasing function; $E' = E\,(E-u) \in (0,1)$, proved with its second-derivative bounds in Lemma 6), and $Z \sim N(0,1)$. Fix $\kappa \in \mathbb{R}$ and write

$$u(z; q, \kappa) = \frac{\kappa - \sqrt{q}\, z}{\sqrt{1-q}}, \qquad 0 < q < 1.$$

Following Ding–Sun and Huang, the replica-symmetric system is

$$P(\psi) = \mathbb{E}\big[\tanh^2(\sqrt{\psi}\, Z)\big], \qquad R(q,\alpha) = \frac{\alpha}{1-q}\,\mathbb{E}\big[E(u(Z))^2\big], \tag{2}$$

$$G(\alpha, q, \psi) = -\frac{\psi(1-q)}{2} + \mathbb{E}\log\big(2\cosh(\sqrt{\psi}\, Z)\big) + \alpha\,\mathbb{E}\log\Psi\big(u(Z)\big), \tag{3}$$

with the fixed-point branch $q = P(\psi)$, $\psi = R(q,\alpha)$ and the threshold $\alpha_\star(\kappa)$ defined as the zero of $G_\star(\alpha) = G(\alpha, q_\star(\alpha), \psi_\star(\alpha))$; we write $(q_0, \psi_0) = (q_\star, \psi_\star)(\alpha_\star(\kappa))$. At $\kappa = 0$, where $u = -\gamma z$ in law with $\gamma = \sqrt{q/(1-q)}$, this is the zero-margin system of [14]. (Display (14) of [10] has $-\alpha\,\mathbb{E}\log\Psi$; the fixed-point equations require the plus sign, which is the sign in [9, eq. (4)].)

Huang's Theorem 3.6 [10] reads: for any $\kappa \in \mathbb{R}$, under his Conditions 1.3, 3.1, 3.3 and 3.4, the probability that some $w \in \{-1,+1\}^N$ satisfies the margin condition for the first $\lceil \alpha N \rceil$ patterns tends to 0 for every $\alpha > \alpha_\star(\kappa)$. Each of the four conditions is defined at every κ and depends on it. Conditions 1, 2, 3 and 4 below are his Conditions 3.1, 3.3, 3.4 and 1.3: Condition 1 is his Condition 3.1 asked on a closed interval, which implies it on the open one; Condition 2 is his Condition 3.3 with F'_{1-q_0} written through the inverse Mills ratio; Condition 3 is his Condition 3.4 with one witness in place of the infimum; and Condition 4 is his Condition 1.3, whose function, his display (8), is written out at general κ in Section 4.1.

Condition 1 (fixed point)**.** *There are an interval $[\alpha_-, \alpha_+]$ and a rectangle $[\alpha_-, \alpha_+] \times [q_{\mathrm{lb}}, q_{\mathrm{ub}}]$ such that for every $\alpha \in [\alpha_-, \alpha_+]$ the map $q \mapsto P(R(q,\alpha))$ has a unique fixed point in $(q_{\mathrm{lb}}, q_{\mathrm{ub}})$, with $\sup_{q \in (q_{\mathrm{lb}}, q_{\mathrm{ub}})} (P \circ R_\alpha)'(q) < 1$, where $R_\alpha = R(\cdot, \alpha)$, and $G_\star(\alpha) = G(\alpha, q_\star(\alpha), \psi_\star(\alpha))$, with $q_\star(\alpha)$ that fixed point and $\psi_\star(\alpha) = R(q_\star(\alpha), \alpha)$, is strictly decreasing on $[\alpha_-, \alpha_+]$ with a unique root there.*

Huang's $\kappa = 0$ instance (his Proposition 3.2) takes an interval of width 10^{-9} about $\alpha_\star(0)$. We call the uniqueness and contraction part the rectangle clause.

Condition 2 (approximate message passing, AMP)**.** *At the fixed point,*

$$\alpha\,\mathbb{E}[\mathrm{sech}^4(\sqrt{\psi_0} Z)]\;\frac{\mathbb{E}[E'(u(Z; q_0, \kappa))^2]}{(1-q_0)^2} \;<\; 1.$$

Condition 3 (local concavity)**.** *Let $A = 1 - q_0$, $F(x) = E\big((\kappa - x)/\sqrt{A}\big)/\sqrt{A}$, $\hat{f}_0 = -F'/(1 + AF')$, $\mu(\hat{z}) = \mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0} Z))^{-1}]$, $d_0 = \alpha_\star \mathbb{E}[F'(\sqrt{q_0} Z)]$ and*

$$\ell(\hat{z}) = \hat{z} - \alpha_\star\,\mathbb{E}\left[\frac{\hat{f}_0(\sqrt{q_0} Z)}{1 + \mu(\hat{z})\hat{f}_0(\sqrt{q_0} Z)}\right] - d_0, \qquad \hat{z} > -1.$$

The condition is that $\ell(\hat{z}) < 0$ for some $\hat{z} > -1$.

This is [10, Condition 3.4], whose function is called λ there; a single witness suffices, since Huang's condition asks that $\inf_{\hat{z}>-1} \ell(\hat{z}) < 0$.

Condition 4 (two-variable function)**.** *$\mathcal{S}_\star(\lambda_1, \lambda_2) \le 0$ for all $(\lambda_1, \lambda_2) \in \mathbb{R}^2$, where $\mathcal{S}_\star$ is the two-variable function of Section 4.1.*

Equality holds at $(1, 0)$ (Section 4.1).

2.1 Two identities

The function G is that of Ding and Sun [9, eq. (4)], who define it at general margin; the two lemmas below record its partial derivatives and the derivative of R at every κ.

Lemma 1 (stationarity)**.** *For all $\alpha > 0$, $\kappa \in \mathbb{R}$, $\psi > 0$ and $0 < q < 1$,*

$$\frac{\partial G}{\partial \psi} = \frac{q - P(\psi)}{2}, \qquad \frac{\partial G}{\partial q} = \frac{\psi - R(q, \alpha)}{2}.$$

In particular the critical points of G are exactly the solutions of Ding and Sun's fixed-point system $q = P(\psi)$, $\psi = R(q, \alpha)$ [9], at every margin.

Proof of Lemma 1. The first identity is Gaussian integration by parts on the $\log\cosh$ term. For the second, differentiating u in q gives

$$u_q = \frac{1}{2\sqrt{1-q}}\Big(-\frac{z}{\sqrt{q}} + \frac{u}{\sqrt{1-q}}\Big), \qquad \frac{\partial}{\partial q}\,\mathbb{E}\log\Psi(u) = -\mathbb{E}\big[E(u)\,u_q\big].$$

Stein's lemma applied to the z part, $\mathbb{E}[zE(u)] = -\sqrt{q/(1-q)}\;\mathbb{E}[E'(u)]$, together with the Mills identity $E'(u) = E(u)^2 - uE(u)$, gives

$$\mathbb{E}[E(u)\,u_q] = \frac{\mathbb{E}[E'(u)] + \mathbb{E}[u\,E(u)]}{2(1-q)} = \frac{\mathbb{E}[E(u)^2]}{2(1-q)} = \frac{R(q,\alpha)}{2\alpha}.$$

Hence $\partial_q G = \psi/2 - \alpha \cdot R/(2\alpha) = (\psi - R)/2$. □

Both derivatives vanish exactly on the fixed-point branch. On a rectangle carrying the bound of Condition 1 the branch $\alpha \mapsto (q_\star, \psi_\star)$ is C^1 (the implicit function theorem applies: $1-(P\circ R_\alpha)'$ is bounded away from zero), and the envelope theorem gives $G'_\star(\alpha) = \mathbb{E}\log\Psi(u) < 0$ strictly ($\Psi < 1$ pointwise), for every κ: the root $\alpha_\star(\kappa)$ is unique on the branch. This is the general-margin form of the monotonicity clause in Condition 1, and it is also what turns the boxes in which Section 3 locates the fixed point (located boxes) into the slabs' $\alpha_\star$ intervals: a certified sign of G at a located box is a certified side of $\alpha_\star$. The branch is increasing in α: P is increasing, since $\tanh^2(\sqrt{\psi}\,|z|)$ increases with ψ for each z; R is linear in α and increasing in q (Lemma 2); so $q'_\star(\alpha) = P'(\psi_\star)\,(R/\alpha)\big/\big(1-(P\circ R_\alpha)'(q_\star)\big) > 0$, and $\psi_\star = R(q_\star, \alpha)$ increases with it. Hence the hull of the boxes located at the two ends of an α interval contains $(q_\star(\alpha), \psi_\star(\alpha))$ for every α in the interval.

Lemma 2 (derivative of R)**.** *For all $\alpha > 0$, $\kappa \in \mathbb{R}$ and $0 < q < 1$,*

$$\frac{\partial R}{\partial q} = \frac{\alpha}{(1-q)^2}\,\mathbb{E}\Big[E'(u)^2 + 2\,E(u)^2 E'(u)\Big] \;>\; 0.$$

Proof. To keep the paper self-contained we derive the formula from (2). Differentiating (2) in q produces $R/(1-q)+\frac{\alpha}{1-q}\mathbb{E}[2EE'u_q]$. Splitting u_q as in Lemma 1 and applying Stein's lemma to $\mathbb{E}[z\,EE'(u)]$ (with $(EE')' = E'^2 + EE''$ and $E'' = E'(E-u) + EE' - E$) turns the cross term into

$$\mathbb{E}[2EE'u_q] = \frac{\mathbb{E}[E'^2 + 2E^2E' - E^2]}{1-q},$$

and the $-\alpha\mathbb{E}[E^2]/(1-q)^2$ piece cancels $R/(1-q)$ exactly. Positivity is immediate since $E' \in (0,1)$. In both proofs differentiation under the integral sign is dominated ($|z\tanh(\sqrt{\psi}z)| \le \sqrt{\psi}\,z^2$, and the q-integrands grow at most quadratically in $|z|$, locally uniformly in the parameters), and Stein's lemma applies because $E(u(z))$ and $(EE')(u(z))$ are absolutely continuous with bounded derivative in z. □

With $B(q) = (1-q)\mathbb{E}[E(u)^2]$ one has $R = \alpha B(q)/(1-q)^2$, and Nakajima's computation of B' [13, eq. (18)] gives at every margin the monotonicity of R that Ding and Sun prove at $\kappa = 0$ [9, Lemma 7.3]. The one-term form is used twice: the integrand is a product of quantities with classical bounds, so its interval enclosure is tight enough for the contraction product $\sup_\psi P' \cdot \sup_q |\partial R/\partial q|$ on each slab (Section 3); and positivity makes $q \mapsto P(R(q))$ increasing, which is what the bisection that locates the fixed point in Section 3 uses.

3 Certificates on a strip

For κ in slabs of width $5\cdot10^{-4}$ covering $[-0.65, 0.19]$, with per-slab parameter rectangles seeded by the continuation, we verify Conditions 1–3 in ball arithmetic (a rectangle is a product of intervals in (α, q), a box a product of intervals in (q_0, ψ_0)) (`python-flint` 0.9.0 on FLINT 3.6.0, the Fast Library for Number Theory, [23], 80-bit working precision, every input a ball covering the whole slab). For a slab, its $\alpha_\star$ interval is the interval $[\alpha_-, \alpha_+]$ of Condition 1 certified for every κ of the slab, and its slab rectangle is $[\alpha_-, \alpha_+] \times [q_{\mathrm{lb}}, q_{\mathrm{ub}}]$.

Theorem 3. *(a) For every $\kappa \in [-0.65, 0.10]$, Conditions 1–3 hold, with the bound of Condition 1 on the slab rectangle.*

(b) For every $\kappa \in (0.10, 0.19]$, Conditions 1–3 hold, with the bound of Condition 1 on the located rectangles of Section 3.4; uniqueness of the fixed point also follows from Nakajima's uniqueness theorem [13, Theorem 1], whose premise is verified on every slab.

(c) In both cases $\alpha_\star(\kappa)$ exists, is unique on the fixed-point branch, and for every slab containing κ it lies in that slab's $\alpha_\star$ interval. If Condition 4 also holds at κ, then for every α above the upper end of that interval the probability that some $w \in \{-1,+1\}^N$ satisfies the margin condition at κ for all $\lceil \alpha N \rceil$ patterns tends to 0 as $N \to \infty$.

The 1680 slabs, each with its κ, α and q intervals, are listed in the verification repository, those with $\kappa \le 0.10$ in the file `verification/results/certified_intervals.csv` and the others in the file `verification/results/certified_intervals_nak.csv`. The proof is given in Sections 3.2–3.4.

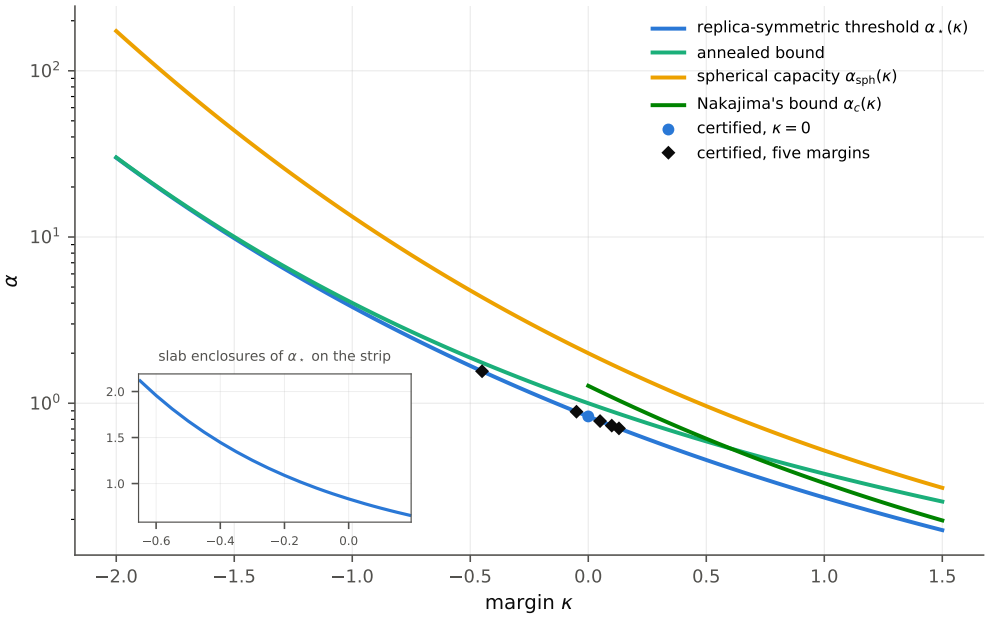


Figure 1: The replica-symmetric threshold $\alpha_\star(\kappa)$ (warm-started continuation from the certified value at $\kappa = 0$), the annealed bound $\log 2/(-\log \Psi(\kappa))$, the spherical capacity $\alpha_{\mathrm{sph}}(\kappa) = 1/\mathbb{E}[(\kappa - Z)_+^2]$ [2, 3], and the bound $\alpha_c(\kappa) = 2/(\pi\, \mathbb{E}[(\kappa - Z)_+^2])$ of Nakajima's uniqueness theorem for $\kappa \geq 0$. The dot is the certified value at $\kappa = 0$ [14] and the diamonds are $\alpha_\star$ at the five margins of the Main Theorem. The inset shows the slabs' $\alpha_\star$ intervals on the strip.

3.1 The replica-symmetric curve

The slab rectangles are seeded by a warm-started continuation in high-precision floating point (Figure 1), which follows the fixed point along $\kappa \in [-2.0, 1.5]$, recording at each of 71 points the threshold, the fixed point (q_0, ψ_0), the derivative at the fixed point $(P \circ R_\alpha)'(q_0)$, the left side of Condition 2 and the local-concavity minimum ℓ_0. Selected values:

κ	$\alpha_\star$	q_0	$(P \circ R_\alpha)'(q_0)$	AMP	ℓ_0
−2.0	29.9921	0.11863	0.387	0.440	−0.147
−1.0	3.79463	0.36519	0.583	0.607	−0.099
−0.5	1.67884	0.47377	0.666	0.589	−0.132
0	0.83308	0.56395	0.744	0.544	−0.191
0.5	0.45566	0.63841	0.810	0.489	−0.271
1.0	0.27070	0.69988	0.863	0.433	−0.359
1.5	0.17245	0.75049	0.903	0.381	−0.442

Conditions 1–3 are satisfied in floating point at every point computed. The derivative at the fixed point rises from 0.387 at $\kappa = -2$ to 0.903 at $\kappa = 1.5$. The minimum ℓ_0 is largest, about -0.094, near $\kappa = -1.3$. On the positive branch $\alpha_\star(\kappa)$ stays below the bound $\alpha_c(\kappa)$ at every grid point (worst ratio 0.874 at $\kappa = 1.5$). Toward $\kappa = -2$, the lower edge of the computed range, the threshold is 29.9921, below the annealed bound $\log 2/(-\log \Psi(-2))$.

3.2 Numerical architecture

The computation rests on three rules.

First, adaptive complex quadrature cannot resolve enclosure width that comes from parameter balls: subdividing the integration variable does not shrink it. So every wide-parameter integral is a real-cell Riemann sum with exact per-cell Gaussian masses. Second, the ball evaluation of φ/Ψ fails on wide arguments in the steep region (the enclosure of Ψ crosses zero), while the ratio is monotone; all factors built from φ/Ψ are therefore enclosed by two thin endpoint evaluations, and the sign-critical sums use the mean-value cell rule (midpoint value plus a derivative correction; on a cell of width h its error is $O(h^2)$ against the plain hull's $O(h)$, with constants set by the derivative enclosures on that cell). Third, expectations that nearly cancel must be evaluated as one correlated integrand, never as separate sums whose parameter widths add. For Condition 3 this is the identity (with $\tau = E'$, $A = 1 - q_0$, and $\beta = \mu(\hat{z}) - A$ for $\mu(\hat{z}) = \mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0}Z))^{-1}]$)

$$\ell(\hat{z}) = \hat{z} + \alpha_\star \mathbb{E}\left[\frac{\beta\,\tau^2}{A(A+\beta\tau)}\right].$$

Indeed $F'(\sqrt{q_0}Z) = -\tau/A$, so $\hat{f}_0 = \tau/(A(1-\tau))$, $\hat{f}_0/(1+\mu\hat{f}_0) = \tau/(A+\beta\tau)$ and $d_0 = -\alpha_\star\mathbb{E}[\tau]/A$, and $\tau/A - \tau/(A+\beta\tau) = \beta\tau^2/(A(A+\beta\tau))$. For fixed (μ, A) the integrand is regular on all of $\tau \in [0,1]$: the map $\tau \mapsto \tau^2/(A+\beta\tau)$ has derivative $\tau(2A+\beta\tau)/(A+\beta\tau)^2$, and $A + \beta\tau$ lies between A and $A + \beta = \mu$, both positive, so $2A + \beta\tau = A + (A + \beta\tau) > 0$. The integrand is therefore monotone in τ, and its absolute value is at most its value at $\tau = 1$, which is $|\beta|/(A\mu)$.

3.3 Uniqueness by contraction

On each of the 1500 slabs of $[-0.65, 0.10]$ five checks are made.

(i) The contraction product $\sup_\psi P' \cdot \sup_q |\partial R/\partial q|$, which bounds $\sup_q (P \circ R_\alpha)'$, is below 1 on the slab rectangle (Lemma 2).

(ii) Existence: the endpoints of the q interval map strictly inward under $P \circ R_\alpha$ on each of eight α sub-intervals.

(iii) The crossing: at the two thin α endpoints a certified Picard iteration locates the fixed point, and $G_\star(\alpha_-) > 0 > G_\star(\alpha_+)$ is certified through a mean-value form that keeps the stationarity cancellations of Lemma 1 explicit. The invariant of the iteration is that the image of any set containing the fixed point contains it.

(iv) Condition 2.

(v) Condition 3 at Huang's witness $\hat{z} = -0.6693$.

Together with the strict decrease of $G_\star$, (i)–(iii) prove Condition 1 on the slab: $\alpha_\star(\kappa)$ exists, is unique, and lies in the slab's $\alpha_\star$ interval for every κ in the slab.

The contraction product (i) is below 1 on each of these 1500 slabs; on $(0.10, 0.19]$ it is not below 1 on every slab (it exceeds 1 near $\kappa = 0.12$), and uniqueness there is treated in Section 3.4. On a single slab the contraction product stays below 1 with slack 0.049 at $\kappa = 0$, growing to 0.198 at $\kappa = -0.65$ and thinning to 0.0083 at $\kappa = 0.10$.

The 1500 slabs are contiguous, cover $[-0.65, 0.10]$, and each satisfies (i)–(iii). The q interval of a slab must cover the drift of the fixed point across the slab's (α, κ) rectangle, which is dominated by $\partial q_\star/\partial\alpha$ times the width of the slab's $\alpha_\star$ interval. At $\kappa = 0.05$, $\partial q_\star/\partial\alpha \approx 0.87$ and, at fixed α, $\partial q_\star/\partial\kappa \approx 1.03$, while along the replica-symmetric curve q_0 changes at about 0.16 per unit of κ.

3.4 Uniqueness on $(0.10, 0.19]$

On each of the 1440 sub-slabs of $(0.10, 0.19]$ (eight of width $6.25 \cdot 10^{-5}$ in each of the 180 slabs covering $[0.10, 0.19]$) the located rectangle is mapped strictly inward at both ends of its $\alpha_\star$ interval and carries a contraction product at most 0.958 (`rect_strip_nak_thin_*.jsonl`, collected with the input hashes bound in `rect_strip_thin_summary.json`). By (2), $R(q, \alpha)$ is α times a positive function of q, and P is increasing, so $P(R(q, \alpha))$ is increasing in α for fixed q; inwardness at the two ends of the $\alpha_\star$ interval therefore gives it at every α between them, and the rectangle clause of Condition 1 holds for every $\kappa \in (0.10, 0.19]$.

The located rectangle of a sub-slab is its $\alpha_\star$ interval times the q interval running from $5 \cdot 10^{-4}$ below the lower to $5 \cdot 10^{-4}$ above the upper of the fixed points located at the two ends of that interval. The fixed points are located by certified sign bisection of the displacement $P(R(q)) - q$, which has certified signs at the ends of the slab's q interval and a single zero between them by Nakajima's theorem [13, Theorem 1]: for $\kappa \geq 0$ and $0 < \alpha < \alpha_c(\kappa) = 2/(\pi\,\mathbb{E}[(\kappa - Z)_+^2])$ the system $q = P(\psi)$, $\psi = R(q, \alpha)$ of (2) has exactly one solution $(q, \psi) \in [0, 1) \times [0, \infty)$. Its premise, $\alpha_+ < \alpha_c(\kappa)$ at the upper end α_+ of the slab's $\alpha_\star$ interval, is certified per slab through the closed form $\mathbb{E}[(\kappa - Z)_+^2] = (1 + \kappa^2)\big(1 - \Psi(\kappa)\big) + \kappa\varphi(\kappa)$; the least certified value of $\alpha_c(\kappa) - \alpha_+$ is 0.292. On a whole slab the enclosure of $P(R(q)) - q$ is wider than its value near the fixed point, so each slab is split into eight sub-slabs whose union covers it.

The slab rectangle itself does not suffice: at $\kappa = 0.13$ its contraction product is 1.32. The slabs of $[0.10, 0.19]$ carry q intervals of width $1.01 \cdot 10^{-2}$, against $7.08 \cdot 10^{-3}$ on the slabs of $[0, 0.10]$, and the product grows with the q width of the rectangle; the drift of the fixed point across a slab's $\alpha_\star$ interval, of width $3.4 \cdot 10^{-3}$, is only about $3.1 \cdot 10^{-3}$ ($\partial q_\star/\partial\alpha \approx 0.9$), while the derivative at the fixed point, $(P \circ R_\alpha)'(q_0)$, stays between 0.758 and 0.770 on $[0.10, 0.19]$.

At the margin $\kappa = 0.13$ of Theorem 4 the rectangle clause is proved as Huang's Proposition 3.2 proves it at $\kappa = 0$, on the rectangle

$$\alpha \in [0.705932217,\ 0.705933898], \qquad q \in [0.5836943,\ 0.5857104].$$

Its q interval is the hull $[0.5846943,\ 0.5847104]$ of the q intervals located at the two α ends, by Picard iteration with mean-value enclosures and Riemann sums of 16000 cells, padded by 10^{-3}. The hull has width $1.6 \cdot 10^{-5}$, so the padding sets the q width of the rectangle. There the contraction product is 0.8435, the map is endpoint-inward at both α ends with slack above $2.3 \cdot 10^{-4}$, the signs of G at the located fixed points are $+1.6 \cdot 10^{-7}$ and $-1.6 \cdot 10^{-6}$, and $\mathbb{E} \log \Psi < 0$ over the rectangle (`rect_cond31_0p13.json`).

Checks (ii), (iv) and (v) hold on each of the 1680 slabs, and checks (i) and (iii) on the 1500 slabs of $[-0.65, 0.10]$. At each of five margins a certified bisection on the sign of G at located boxes encloses $\alpha_\star$ in an interval of width about $2 \cdot 10^{-6}$; these $\alpha_\star$ intervals are the intervals of Theorem 4.

Proof of Theorem 3. (a) The 1500 slabs of $[-0.65, 0.10]$ are contiguous and cover it. On each, checks (i)–(iii) of Section 3.3 and the strict decrease of $G_\star$ give Condition 1 with the bound

on the slab rectangle, and checks (iv) and (v) give Conditions 2 and 3. (b) The 180 slabs of $[0.10, 0.19]$ are contiguous and cover it. On each, the premise of Nakajima's theorem, the existence check (ii), the location of the fixed points at the two α ends by sign bisection with the signs of G, and checks (iv) and (v) are certified; the contraction product is below 1 on the located rectangle of each of its eight sub-slabs (Section 3.4). This gives Conditions 1–3, with the bound of Condition 1 on the located rectangles, and Nakajima's theorem gives uniqueness of the fixed point. (c) On a slab, Condition 1 gives existence and uniqueness of $\alpha_\star(\kappa)$ on the fixed-point branch and places it in the slab's $\alpha_\star$ interval. If Condition 4 also holds at κ, all four of Huang's conditions hold there, and [10, Theorem 3.6] gives the limit. □

4 The two-variable condition

Theorem 4. *Let $\kappa \in \{-0.45, -0.05, 0.05, 0.0995, 0.13\}$. Then Conditions 1–4 hold at κ, $\alpha_\star(\kappa)$ lies in the corresponding interval*

$$\begin{aligned}
&\alpha_\star(-0.45) \in [1.557652103,\ 1.557654133], \quad \alpha_\star(-0.05) \in [0.889408031,\ 0.889409783],\\
&\alpha_\star(0.05) \in [0.781073068,\ 0.781074776], \quad \alpha_\star(0.0995) \in [0.733478513,\ 0.733480205],\\
&\alpha_\star(0.13) \in [0.705932217,\ 0.705933898],
\end{aligned}$$

and, by [10, Theorem 3.6], for every α above that interval

$$\lim_{N\to\infty} \mathbb{P}\Big(\exists\, w \in \{-1,+1\}^N :\ \frac{\langle g_a, w\rangle}{\sqrt{N}} \geq \kappa \text{ for all } a \leq \lceil \alpha N \rceil\Big) = 0.$$

4.1 The function at general margin

Huang's functional at margin κ, his display (8) with $\eta = F(\hat{H})$ written for his $F_{1-q_0}(\hat{H})$, is

$$\mathcal{S}(\Lambda, s) = \frac{s^2\psi_0}{2} + \mathrm{ent}(\Lambda) + \alpha\, \mathbb{E} \log \Psi\left(\frac{\kappa - (a_2/q_0)\hat{H} - (a_1/\psi_0)\eta}{D} + s\eta \right),$$

where $\dot{H} \sim N(0, \psi_0)$ and $\hat{H} \sim N(0, q_0)$ are independent, $M = \tanh \dot{H}$ (Huang's notation), the profile Λ is a function of $\dot{H}$ with $|\Lambda| \leq 1$, $\mathrm{ent}(\Lambda) = \mathbb{E}\, h\big(\frac{1+\Lambda}{2}\big)$ with $h(p) = -p\log p - (1-p)\log(1-p)$, $\eta = E(u)/\sqrt{1-q_0}$ with $u = (\kappa - \hat{H})/\sqrt{1-q_0}$ (the margin enters through η and through the additive κ in the $\log\Psi$ argument), $a_1 = \mathbb{E}[\dot{H}\Lambda]$, $a_2 = \mathbb{E}[M\Lambda]$, $D = D(a) = \sqrt{1 - a_2^2/q_0}$, and $\mathcal{S}_\star(\lambda_1, \lambda_2) = \inf_{s\geq 0} \mathcal{S}(\tanh(\lambda_1\dot{H} + \lambda_2 M), s)$. We call $\mathrm{ent}(\Lambda)$ the entropy term and the term $\alpha\, \mathbb{E}\log\Psi(\cdot)$ the constraint term. Condition 4 asks $\mathcal{S}_\star \leq 0$ everywhere.

The moment reduction [14, Lemma 5.1] holds at every κ with the same proof, because the margin enters only the constraint term. The profile enters only through the moment pair $a(\Lambda) = (\mathbb{E}[\dot{H}\Lambda], \mathbb{E}[M\Lambda])$, which ranges over a compact convex body K (the moment body of $(\dot{H}, M)$), and on K the entropy term is bounded through $H(a) = \max\{\mathrm{ent}(\Lambda) : \mathbb{E}[\dot{H}\Lambda] = a_1,\ \mathbb{E}[M\Lambda] = a_2\}$ by convex duality, $H(a) \leq \Phi(b) - b\cdot a$ for every dual point b, with $\Phi(b) = \mathbb{E}\log 2\cosh(b_1\dot{H} + b_2 M)$ free of κ. With

$$\mathcal{F}(a, s) = \frac{s^2\psi_0}{2} + H(a) + \alpha\, \mathbb{E}\log\Psi\left(\frac{\kappa - (a_2/q_0)\hat{H} - (a_1/\psi_0)\eta}{D(a)} + s\eta \right), \qquad \bar{\mathcal{S}}(a) = \inf_{s\geq 0} \mathcal{F}(a, s),$$

the lemma gives

$$\mathcal{S}_\star(\lambda) \le \bar{\mathcal{S}}\big(a(\tanh(\lambda_1 \dot{H} + \lambda_2 M))\big), \qquad \lambda \in \mathbb{R}^2.$$

The certificates bound $\mathcal{F}$ with $H(a)$ replaced by $\Phi(b) - b \cdot a$ at the dual point b of each cell. The maximizing profile $\tanh \dot{H}$, that is $\lambda = (1, 0)$, has moment pair $a^\star = (\psi_0(1 - q_0),\, q_0)$ for every κ: this is the identity $\nabla\Phi(1, 0) = (\mathbb{E}[\dot{H}M], \mathbb{E}[M^2]) = (\psi_0(1 - q_0), q_0)$ (Gaussian integration by parts and the fixed-point equations), and [10, Lemma 7.15] gives $\mathcal{S}_\star = 0$ there with optimal tilt $s_0 = \sqrt{1 - q_0}$, both in closed form. And on a region close to the half-plane $\lambda_1 < 0$ the s-infimum is $-\infty$: $\alpha\, \mathbb{E}[\eta^2] = \psi_0$ exactly at the fixed point, so the quadratic tilt coefficient cancels, and wherever the surviving linear coefficient of the tilt is nonpositive $\mathcal{S}(\Lambda, s) \to -\infty$ as $s \to \infty$ (in floating point, the boundary of that region at $\kappa = 0.05$ crosses $\lambda_1 = 0.04$ on the $\lambda_2 = 0$ axis and bends left as $|\lambda_2|$ grows). Condition 4 takes the infimum over all $s \ge 0$, so a negative value at one finite tilt is a witness, and the certificates of Section 4.2 exhibit one cell by cell.

In floating point the function is nearly the same at every κ of the curve: a single peak at $(1, 0)$, off-center values below $-3 \cdot 10^{-3}$, negative far rays, eigenvalues near -0.22 and -0.0045 of the Hessian of $\mathcal{S}_\star$ in λ at $(1, 0)$ (finite differences), distance near 0.017 from the boundary of K; the eigenvalue nearer 0 is -0.0013 at $\kappa = -1.5$.

4.2 Certifying the two-variable condition

The zero-margin argument has four certificates: Region II, a sweep of the moment body outside an exclusion region, the exclusion region being a product of two intervals about $a^\star$ in the moment plane that contains the star, the set of points $a^\star + tv$ with $0 \le t \le T(v)$ for the certified radius $T(v)$ of each unit direction v; Region I, ray certificates on the star, one for each radial band and angular cell; stage 2, a sweep of the exclusion region minus the star; and the star interior, a certificate that the star lies strictly inside the moment body. At $\kappa \ne 0$ all four meet the same difficulty. At zero margin the parameters (q_0, ψ_0) are certified to a box of width $8 \cdot 10^{-10}$ in q_0 and $1.2 \cdot 10^{-8}$ in ψ_0, so the moment pair $a^\star$ can be treated as an exact point; at $\kappa = 0.05$ the hull of the boxes located at the two ends of the $\alpha_\star$ interval has width $1.5 \cdot 10^{-6}$ in q_0 and $9.2 \cdot 10^{-5}$ in ψ_0 (at each end q_0 is located to $1.5 \cdot 10^{-9}$ and ψ_0 is enclosed as $R(q_0, \alpha)$ by a Riemann sum of 4000 cells, to width $6.9 \cdot 10^{-5}$). The certificates of this section use this hull; the Picard box of Section 4.4 is narrower in ψ_0 and wider in q_0, and both contain the fixed point for every α of the $\alpha_\star$ interval; their intersection is narrower in ψ_0 by a factor of 1.8, and the certificates do not need it. The width of the hull enters every plain evaluation of Φ at about $2 \cdot 10^{-5}$, much wider than the differences of order 10^{-7} whose signs place the dual points. Four changes make the four certificates apply at $\kappa \ne 0$: $a^\star$ computed from the hull, differences of Φ written against $b_0 = (1, 0)$, bounds on the inverse Mills ratio that keep the constraint kernels finite where its argument is large, and one tilt slope per angular sector on the rays.

The moment pair from the hull. Every certificate that references $a^\star$ or the optimal tilt s_0 uses them as the functions $(\,\psi_0(1 - q_0),\, q_0\,)$ and $\sqrt{1 - q_0}$ of (q_0, ψ_0) on the hull, never as stored decimals; the enclosure of $a^\star$ is the image of the hull under the first of these maps. The true $a^\star$, at the true parameters, is therefore covered exactly, and the identities at $a^\star$ ($\mathcal{F}(a^\star, s_0) = \mathcal{S}(\tanh \dot{H}, s_0) = 0$, and $\nabla_a \mathcal{F} = 0$, $\partial_s \mathcal{F} = 0$ by Lemma 5) hold on the certified rays with no enlargement of the enclosure. Certificates evaluated over the enclosure of $a^\star$ cover in particular the rays from $a^\star$ at the fixed point.

Lemma 5 (gradient at the moment pair)**.** *Let* (q_0, ψ_0) *satisfy* $q_0 = P(\psi_0)$ *and* $\psi_0 = R(q_0, \alpha)$. *With* $a^\star = (\psi_0(1-q_0), q_0)$ *and* $s_0 = \sqrt{1-q_0}$, $\nabla_a \mathcal{F}(a^\star, s_0) = 0$ *and* $\partial_s \mathcal{F}(a^\star, s_0) = 0$, *at every* κ.

Proof. Since $H(a) = \min_b\{\Phi(b) - b \cdot a\}$ and $a^\star = \nabla\Phi(b_0)$ with $b_0 = (1, 0)$, $\nabla H(a^\star) = -b_0$. Write $X(a) = \big(\kappa - (a_2/q_0)\hat{H} - (a_1/\psi_0)\eta\big)/D(a)$, so that $X(a^\star) + s_0\eta = u$ and $D(a^\star) = \sqrt{1-q_0}$. At $a^\star$, $\partial X/\partial a_1 = -\eta/(\psi_0\sqrt{1-q_0})$ and $\partial X/\partial a_2 = Q/\sqrt{1-q_0}$, with $Q = -\hat{H}/q_0 + \big(\kappa - \hat{H} - (1-q_0)\eta\big)/(1-q_0)$. Since $(\log\Psi)' = -E$ and $E(u) = \sqrt{1-q_0}\,\eta$,

$$\partial_{a_1}\mathcal{F} = -1 + \frac{\alpha\,\mathbb{E}[\eta^2]}{\psi_0}, \qquad \partial_{a_2}\mathcal{F} = -\alpha\,\mathbb{E}[\eta Q], \qquad \partial_s\mathcal{F} = s_0\big(\psi_0 - \alpha\,\mathbb{E}[\eta^2]\big).$$

The first and the third vanish because $\alpha\,\mathbb{E}[\eta^2] = \psi_0$. For the second, Stein's lemma gives $\mathbb{E}[\eta\hat{H}] = -q_0\mathbb{E}[E'(u)]/(1-q_0)$, and $\kappa - \hat{H} = \sqrt{1-q_0}\,u$, so

$$\mathbb{E}[\eta Q] = \frac{\mathbb{E}[E'(u)] + \mathbb{E}[uE(u)] - \mathbb{E}[E(u)^2]}{1-q_0} = 0$$

by the identity $E' = E^2 - uE$. □

The ray argument. This is the ray part of [14, Lemma 6.1], restated at general κ; the two facts it uses at the origin of a ray hold at every κ by Lemma 5. For a unit direction v and a slope $c \in \mathbb{R}$ put $\chi_{v,c}(t) = \mathcal{F}(a^\star + tv,\, s_0 + ct)$, with $\mathcal{F}$ as in Section 4, so that $\bar{\mathcal{S}}(a^\star + tv) \le \chi_{v,c}(t)$ whenever $s_0 + ct \ge 0$. At the fixed point (q_0, ψ_0) of $\alpha = \alpha_\star(\kappa)$, $\chi_{v,c}(0) = 0$, and $\chi'_{v,c}(0) = 0$ for every c by Lemma 5. If $\chi''_{v,c} \le 0$ on $[0, T(v)]$ then $\chi_{v,c} \le 0$ there, hence $\bar{\mathcal{S}} \le 0$ on the segment. The chain needs one slope per ray: the second-derivative bound on a radial band $[t_0, t_1]$ with $t_0 > 0$ says nothing about χ itself unless χ and χ' are controlled at t_0, and those are controlled only through the same function on $[0, t_0]$. The certificates therefore fix the slope by direction alone: the circle is cut into 768 equal sectors, each sector carries one slope (the optimal tilt slope at its midpoint, rounded to six decimals), every angular cell of every radial band lies inside one sector and uses that sector's slope, and a direction on a sector edge is covered by the cells of either neighbor, each with its own fixed slope. The cells of each band tile its angle range, and the main and supplementary certificates use the same slope table.

Differences against b_0. The dual localization of [14, Lemma 6.1], which keeps the optimal dual point $\lambda(a)$ inside a certified region of the dual plane (its dual enclosure) so that the entropy Hessian can be bounded there, needs signs of differences $\Phi(\lambda) - \Phi(\lambda') - (\lambda - \lambda') \cdot a$ for pairs of dual points λ, λ' and moment pairs a near $a^\star$. Evaluated as differences of plain Φ enclosures these carry the width of the hull twice. Written against $b_0 = (1, 0)$, with $\Delta = (\lambda_1 - 1)\dot{H} + \lambda_2 M$, the integrand satisfies

$$\log\frac{\cosh(\dot{H} + \Delta)}{\cosh\dot{H}} - (\lambda - b_0) \cdot \nabla_b \log 2\cosh(b \cdot (\dot{H}, M))\big|_{b_0} = \log\big(\cosh\Delta + M\sinh\Delta\big) - M\Delta,$$

because the gradient of $b \mapsto \log 2\cosh(b \cdot (\dot{H}, M))$ at b_0 is $M\,(\dot{H}, M)$, whose inner product with $\lambda - b_0$ is $M\Delta$. Every factor is a small ball with thin coefficients, and the enclosure of the right side over the box has width proportional to $|\lambda - b_0|$. At distance 0.01 from b_0, where plain differences of enclosures fail, it is 57 to 251 times narrower, depending on the direction, and at b_0 it is exactly $\{0\}$. Differences between arbitrary points are then differences of two

such enclosures, and the gradient term the localization needs is the same identity for $\nabla\Phi$: $\nabla\Phi(\lambda) - \nabla\Phi(b_0)$ has the exact per-node form $(\dot{H}, M)\,(1 - M^2)\tanh\Delta/(1 + M\tanh\Delta)$.

The factor $\cosh\Delta + M\sinh\Delta$ itself has two exact evaluations with disjoint stability regimes: the cosh/sinh form is exact at $\Delta = 0$ (which the differences of order 10^{-7} in the bands nearest $a^\star$ require) but subtracts two $e^{|\Delta|}$-sized balls when $M \approx \pm 1$ and $|\Delta|$ is large, where the enclosure of the factor contains 0 and its logarithm is not enclosed; the split $e^{\Delta}(1+M)/2 + e^{-\Delta}(1-M)/2$ has both terms nonnegative on the real axis but carries first-order M-width near $\Delta = 0$. The first form is used at nodes with $|\Delta| \le 1$, the second at the others; both forms are identities, so the choice affects tightness only.

Kernels at large arguments. The ray certificates use second derivatives of the constraint term, whose kernels involve $E'(V) = E(V)(E(V) - V)$ and $E''(V)$, with E the inverse Mills ratio of Section 2, evaluated at arguments V that reach about 20 on far quadrature cells; there $E(V) - V$ is a difference of two large balls and the enclosures of the kernels are useless. The bounds of the following lemma, applied as intersections with the computed balls, remove the failure; the intersection of an enclosure with a proven bound is again an enclosure.

Lemma 6 (bounds on the inverse Mills ratio)**.** *Let E be the inverse Mills ratio and $\sigma = E - V$. For every $V \in \mathbb{R}$,*

$$\sigma > 0 \quad \Big(\textit{and } \sigma < \frac{1}{V} \textit{ for } V > 0\Big), \qquad E' = E\sigma \in (0,1), \qquad 0 < E'' < \sigma.$$

Proof. For $V \le 0$, $\sigma > 0$ because $E > 0$. For $V > 0$ the bounds $V < E(V) < V + 1/V$ are Gordon's inequalities [7], which give $0 < \sigma < 1/V$. The bounds $0 < E' < 1$ are due to Sampford [8]; here $E' = E^2 - VE = E\sigma$, and $1 - E' = \mathrm{Var}(Z \mid Z > V)$ by the conditional moments $\mathbb{E}[Z \mid Z > V] = E$ and $\mathbb{E}[Z^2 \mid Z > V] = 1 + VE$, which are used again below. The upper bound on E'' is the identity $E'' - \sigma = (E' - 1)(E + \sigma)$ with $E' < 1$ and $E + \sigma > 0$.

Positivity of E'' is the convexity of the inverse Mills ratio, proved by Sampford [8]; to keep the paper self-contained we give a short proof, which compares the residual law of Z beyond V with the exponential law of the same mean. Differentiating E' gives $E'' = E\,[\,\sigma(E+\sigma) - 1\,]$, and for W distributed as $Z - V$ conditioned on $\{Z > V\}$ the moment formulas above read $\mathbb{E}W = \sigma$ and $\mathbb{E}[W^2] = 1 - VE + V^2$, whence $\sigma(E+\sigma) - 1 = 2(\mathbb{E}W)^2 - \mathbb{E}[W^2]$: the claim is that this residual law has coefficient of variation below one. Its survival function is $\mathbb{P}(W > w) = \Psi(V + w)/\Psi(V) = e^{-\Gamma(w)}$ with $\Gamma(w) = \int_0^w E(V + x)\,dx$ convex, since $E' > 0$. Set $\theta = 1/\sigma$, so the rate-θ exponential has the same mean, and $J(w) = \Gamma(w) - \theta w$. Then J is convex with $J(0) = 0$, $J'(0) = E - 1/\sigma < 0$ (this is $E\sigma = E' < 1$ again), and $J(w) \to \infty$, so $J < 0$ exactly on some interval $(0, w_0)$ and $J > 0$ past it. Hence $\mathbb{P}(W > w) - e^{-\theta w} = e^{-\theta w}(e^{-J(w)} - 1)$ is positive before w_0 and negative after, while $\int_0^\infty (\mathbb{P}(W > w) - e^{-\theta w})\,dw = \sigma - \sigma = 0$. Therefore

$$\tfrac{1}{2}\,\mathbb{E}[W^2] - \sigma^2 = \int_0^\infty w\,\big(\mathbb{P}(W > w) - e^{-\theta w}\big)\,dw = \int_0^\infty (w - w_0)\big(\mathbb{P}(W > w) - e^{-\theta w}\big)\,dw < 0,$$

the last integrand being nonpositive and not almost everywhere zero. So $\mathbb{E}[W^2] < 2(\mathbb{E}W)^2$, that is $E'' > 0$, at every real V. □

With these four changes the four certificates hold at each of the five margins. The counts at $\kappa = 0.05$ follow; Table 1 gives the other margins. At $\kappa = 0.05$, on that hull and the $\alpha_\star$ interval of Theorem 4:

κ	$\alpha_\star(\kappa)$ interval	Region II	Region I	stage 2	star interior
0.05	[0.781073068, 0.781074776]	855	1404 (+12)	1363	0.01481 / 0.01209
−0.45	[1.557652103, 1.557654133]	656	1404 (+12)	1196	0.01480 / 0.01207
−0.05	[0.889408031, 0.889409783]	867	1404 (+12)	1282	0.01527 / 0.01209
0.0995	[0.733478513, 0.733480205]	904	1404 (+12)	1390	0.01459 / 0.01209
0.13	[0.705932217, 0.705933898]	874	1404 (+12)	1422	0.01446 / 0.01210

Table 1: Certified leaf and band counts per certificate. In the Region I column, (+12) marks the twelve supplementary arcs that follow the bands; the stage-2 lattice of every margin uses them. The star-interior column gives the constructive moment-body inradius against the required star radius, both exact-arithmetic bounds.

(i) *Region II*: the 1296 cells of a 48×27 grid on $[-1.351, 1.351]\times[-0.76, 0.76]$, which contains K; cells inside the exclusion region or outside K need no bound, and the others are bisected until the duality bound certifies each leaf, 855 certified leaves in all.

(ii) *Region I*: the ray certificates over the star, 1404 bands from $a^\star$; the certified band values lie between $-1.6\cdot10^{2}$ and $-7.5\cdot10^{-5}$. The main radius is 0.012 within 0.16 radians of the weak direction, the line through $a^\star$ along a unit eigenvector of the Hessian of $\bar{\mathcal{S}}$ at $a^\star$ for its eigenvalue nearest 0 (computed in floating point; the choice of the angular ranges does not enter the proof), 0.008 out to 0.45 radians and 0.005 beyond. Twelve supplementary arcs, angular ranges 0.14 to 0.19 radians off the weak direction, certify radius 0.0105 there (certified values $-3.5\cdot10^{-3}$ to -1.9), as the stage-2 leaves near those arcs require.

(iii) *Stage 2*: the exclusion region minus the star, 1363 leaves. Here $T(v)$ is the larger of the main radius at v and 0.0105 if v lies in a supplementary arc, a piecewise-constant function of the angle.

(iv) *Star interior*: the star-interior certificate of [14, Lemma 5.2], run over the enclosure of $a^\star$, places the star strictly inside the moment body: constructive inradius 0.01481 against required 0.01209.

Together: $\mathcal{S}_\star \le 0$ on the whole plane at $\kappa = 0.05$. All four of Huang's conditions hold there, and his theorem gives the unconditional bound.

4.3 Further margins

Huang's four conditions are statements at one margin, and the certificates of Sections 4.1–4.2 run unchanged at any κ of the strip, with the boxes of (q_0, ψ_0) located at both ends of the $\alpha_\star$ interval of that margin. Besides $\kappa = 0.05$, Condition 4 holds at $\kappa = -0.45$, -0.05, 0.0995 and 0.13. At the four margins with $\kappa \le 0.10$ the uniqueness of the fixed point comes from the contraction certificates of Section 3 alone; at $\kappa = 0.13$ Nakajima's uniqueness theorem gives it as well. Table 1 lists the certified leaves and bands at the five margins.

For each band, the ray-certificate file of every margin carries a Loewner majorant $\bar{B}$, an interval matrix with $\nabla^2\Phi(\lambda) \preceq \bar{B}$ in the Loewner order (the difference is positive semidefinite) for every λ in the band's dual enclosure, so that the Hessian of the entropy term is at most

$-\bar{B}^{-1}$ there. Every band's $\bar{B}$ is positive definite, a fact checked in exact rational arithmetic (Section 4.5).

4.3.1 Intervals of κ

For Conditions 1–3 the strip certificates already give statements on intervals of κ: each slab is certified with every input enclosed over the whole slab. Condition 4 is different. Over a slab every certificate must hold with (α, q_0, ψ_0) replaced by their hulls over the slab. The slabs' $\alpha_\star$ intervals of Section 3 are between $3.4 \cdot 10^{-3}$ and $4.7 \cdot 10^{-3}$ wide, and across such an interval the fixed point moves in q_0 at rate about 0.9, so the hulls are about $3 \cdot 10^{-3}$ wide in q_0; the induced enclosure of $a^\star$ has radius at least a few times 10^{-3}, and at that radius the Region II cell enclosures are about ± 0.2, while the thinnest Region II cells have slack $5 \cdot 10^{-4}$ and the thinnest Region I bands between $1.1 \cdot 10^{-5}$ and $7.5 \cdot 10^{-5}$. The cell enclosures would have to come down by a factor of about 400, and thinner slabs reduce the hulls only as far as the slabs' $\alpha_\star$ intervals narrow, which is limited by the enclosure of $G_\star$.

The smaller eigenvalue of the Hessian of $\mathcal{S}_\star$ at $(1, 0)$ is -0.0013 at $\kappa = -1.5$ (Section 4.1), and the minimum ℓ_0 is largest, about -0.094, near $\kappa = -1.3$; whether Condition 4 holds at every negative margin is open.

4.4 The maximizer and its Hessian

At $\kappa \in \{-0.45, -0.05, 0, 0.05, 0.0995, 0.13\}$ we show negative definiteness of the Hessian

$$\mathcal{H} = \nabla^2_\lambda \, \mathcal{S}\big(\tanh(\lambda_1 \dot{H} + \lambda_2 M), s_0\big)\Big|_{\lambda=(1,0)}$$

of Huang's functional $\mathcal{S}$ of Section 4.1, the tilt held at its optimal value $s_0 = \sqrt{1-q_0}$; at $\kappa = 0$ this is [10, Claim 2.6], verified in his Appendix B.3. It is the local form of Condition 4 at $(1, 0)$; the certificates of Section 4.2 do not use it. The second derivative in the tilt is positive: it is $\psi_0 - \alpha \mathbb{E}[E' \eta^2]$, with $\eta = E(u)/\sqrt{1-q_0}$ and $0 < E' < 1$ (Lemma 6) and $\alpha \mathbb{E}[\eta^2] = \psi_0$. Minimizing over the tilt therefore subtracts a positive semidefinite matrix (a Schur complement), so the Hessian of $\mathcal{S}_\star$ at $(1, 0)$ is bounded above by $\mathcal{H}$ in the Loewner order and is negative definite whenever $\mathcal{H}$ is. With the tilt held at s_0, $\mathcal{S}(\tanh(\lambda_1 \dot{H} + \lambda_2 M), s_0) = H(a(\lambda)) + \mathcal{C}(a(\lambda)) + s_0^2 \psi_0/2$, where $a(\lambda) = \nabla \Phi(\lambda)$, $\mathcal{C}(a) = \alpha \, \mathbb{E} \log \Psi(X(a) + s_0 \eta)$ and H, Φ, X are as in Section 4.1 and Lemma 5. At $\lambda = b_0 = (1, 0)$ the gradient of $H + \mathcal{C}$ in a vanishes (Lemma 5), the derivative of $a(\lambda)$ is $\nabla^2 \Phi(b_0)$ and $\nabla^2 H(a^\star) = -\nabla^2 \Phi(b_0)^{-1}$, so the chain rule gives at every κ the quadratic form

$$\langle \xi, \mathcal{H} \xi \rangle = -(I_1 \xi_1^2 + 2 I_2 \xi_1 \xi_2 + I_3 \xi_2^2) + C_1 y_1^2 + C_2 y_1 y_2 + C_3 y_2^2,$$

where $y = \nabla^2 \Phi(b_0) \xi$, that is $y_1 = I_1 \xi_1 + I_2 \xi_2$ and $y_2 = I_2 \xi_1 + I_3 \xi_2$, with the entries of $\nabla^2 \Phi(b_0)$ ($I_1 = \psi_0(1-q_0) - 2\psi_0^2(1 - 4q_0 + 3p_4)$, $I_2 = \psi_0(1 - 4q_0 + 3p_4)$, $I_3 = q_0 - p_4$, $p_4 = \mathbb{E} \tanh^4(\sqrt{\psi_0} Z)$), and $C_1 = \partial_{a_1}^2 \mathcal{C}$, $C_2 = 2 \, \partial_{a_1} \partial_{a_2} \mathcal{C}$, $C_3 = \partial_{a_2}^2 \mathcal{C}$ at $a^\star$. Put $\hat{H} = \sqrt{q_0} \, Z$, $u = (\kappa - \hat{H})/\sqrt{1-q_0}$, $\eta = E(u)/\sqrt{1-q_0}$ and $F' = -E'(u)/(1-q_0)$. At $\kappa = 0$ Huang's closed forms use the two identities $\alpha \, \mathbb{E}[\eta^2] = \psi_0$ and $\alpha \, \mathbb{E}[\eta \hat{H}] = -q_0(1-q_0)\psi_0$. The first holds at every margin, since it is the second equation of (2); the second fails at $\kappa \neq 0$ (at $\kappa = 0.05$ the two sides differ by 0.0335). From $(\log \Psi)' = -E$, $(\log \Psi)'' = -E'$ and the first and second derivatives of X, with

$Q = -\hat{H}/q_0 + \big(\kappa - \hat{H} - (1-q_0)\eta\big)/(1-q_0)$, every expectation over the single Gaussian $\hat{H}$,

$$C_1 = \frac{\alpha}{\psi_0^2}\,\mathbb{E}[F'\eta^2], \qquad C_2 = -\frac{2\alpha}{\psi_0}\,\mathbb{E}[F'\eta Q] + \frac{2}{1-q_0},$$

$$C_3 = \alpha\,\mathbb{E}[F'Q^2] + \frac{2\alpha}{q_0(1-q_0)}\,\mathbb{E}[\eta\hat{H}] - \alpha\Big(\frac{3}{(1-q_0)^2} + \frac{1}{q_0(1-q_0)}\Big)\mathbb{E}\big[\eta\,(\kappa - \hat{H} - (1-q_0)\eta)\big].$$

The certified evaluation encloses each expectation by the mean-value cell rule with explicit quadratic tail majorants (η grows at most linearly on the left tail, F' is bounded by $1/(1-q_0)$ in magnitude, and every integrand is degree at most two in $|z|$), and the near-cancelling prior moments are evaluated as single correlated integrands. The certified enclosures at the six margins are:

$$\begin{array}{lll}
\kappa = -0.45: & \mathcal{H}_{11} \le -0.0647, & \det\mathcal{H} \ge 2.9\cdot 10^{-4},\\
\kappa = -0.05: & \mathcal{H}_{11} \le -0.0471, & \det\mathcal{H} \ge 1.6\cdot 10^{-4},\\
\kappa = 0: & \mathcal{H}_{11} \le -0.0451, & \det\mathcal{H} \ge 1.8\cdot 10^{-4},\\
\kappa = 0.05: & \mathcal{H}_{11} \le -0.0424, & \det\mathcal{H} \ge 1.2\cdot 10^{-4},\\
\kappa = 0.0995: & \mathcal{H}_{11} \le -0.0400, & \det\mathcal{H} \ge 1.0\cdot 10^{-4},\\
\kappa = 0.13: & \mathcal{H}_{11} \le -0.0385, & \det\mathcal{H} \ge 8.8\cdot 10^{-5}.
\end{array}$$

The bound on $|\mathcal{H}_{11}|$ decreases as κ increases.

The C_j move with q at rate about 10 and $\det\mathcal{H}$ is of order 10^{-4}, so the slab rectangles are too wide. Iterating $q \mapsto P(R(q,\alpha))$ over the whole $\alpha_\star$ interval gives a box, the Picard box, that contains the fixed point for every α in it ($3.6\cdot 10^{-6}$ in q_0, $5.2\cdot 10^{-5}$ in ψ_0 at $\kappa = 0.05$). On the Picard box $\mathcal{H}_{11} < 0$ and $\det\mathcal{H} > 0$ in ball arithmetic, with the near-cancellations $1 - 4q_0 + 3p_4$ and $q_0 - p_4$ evaluated as single correlated integrands and the five constraint-side integrals by mean-value Riemann sums with elementary derivatives of φ/Ψ. The same chain certifies at $\kappa = 0$ on the parameter rectangle of [9, Section 7], where the C_j enclosures lie inside Huang's certified intervals, and at -0.05, 0.0995, -0.45 (where $C_1 = -0.855$, against -0.718 at zero margin) and 0.13.

Along the continuation $\mathcal{H}$ stays negative definite at least up to $\kappa = 1.15$ and loses definiteness inside $(1.1812, 1.1844)$, while the Hessian of $\mathcal{S}_\star$ remains negative out to at least $\kappa = 1.5$, as the Schur relation allows, so past $\kappa \approx 1.18$ a local argument must re-optimize the tilt, as the ray certificates of Section 4.2 already do.

4.5 The certificates and their exact part

Every inequality of Theorems 4 and 3 is decided by a comparison of arb balls (python-flint 0.9.0 on FLINT 3.6.0), built from the primitives of [14]: monotone endpoint evaluations, mean-value cells, correlated integrands and tail bounds. The identities the certificates use are Lemmas 1, 2 and 6, Lemma 5, the formula for ℓ (Section 3.2) and the identity written against b_0 (Section 4.2).

The certificates of Condition 4 record every sign-critical quantity as an outward-rounded decimal in canonical JSON, bound by SHA-256 hashes; the stage-2 file binds the hash of the Region I file it reads. The steps that use only these recorded rationals are restated over the integers and checked by the kernel of Lean 4 [24] (core library, v4.35.0): the inequality chain of the star interior, including the positivity of the determinant of the 2×2 matrix whose parallelogram gives the inradius, the positive definiteness of the Loewner majorants

$\bar{B}$ of Section 4.3 at all five margins, and the gap-free tilings of the Region II and stage-2 grids and of the 1404-band Region I schedule. Two global clauses on the band geometry, that the innermost ring closes the full circle and that each level's angular support lies inside the previous level's, together make every radial chain contiguous out to the star radius. They are checked in exact rational arithmetic on the band endpoints recorded in the Region I file by `portable_check.py`, which uses only the Python standard library; they and the analytic lemmas are not part of the statement checked in Lean.

The enclosures behind Theorem 3 are not restated in exact arithmetic; that theorem rests on the ball-arithmetic computation alone.

Proof of Theorem 4. Fix one of the five margins. Conditions 1–3 hold at κ by Theorem 3, and the certified bisection of Section 3.4 encloses $\alpha_\star(\kappa)$ in the stated interval. For Condition 4, the moment reduction of Section 4.1 gives $\mathcal{S}_\star(\lambda) \le \bar{\mathcal{S}}(a)$ for the moment pair $a \in K$ of the profile, so it suffices that $\bar{\mathcal{S}} \le 0$ on K. This holds outside the exclusion region by Region II, on the exclusion region minus the star by stage 2, and on the star by Region I and the ray argument, which needs the star inside K; the star-interior certificate gives that. The four certificates hold at $\kappa = 0.05$ with the counts of Section 4.2 and at the other four margins with those of Table 1. With the four conditions, [10, Theorem 3.6] gives the limit. □

Condition 4 holds at the five margins of Theorem 4 and, by [14], at $\kappa = 0$. At every other κ of the strip, Theorem 3(c) reduces the upper bound to Condition 4 at that κ.

Remark 7. For binary weights the Euclidean norm is fixed at $\sqrt{N}$, so the margin condition is a robustness certificate: if $y_a\langle w, x_a\rangle/\sqrt{N} \ge \kappa$, the label y_a is unchanged under every perturbation of x_a of Euclidean norm below κ. With standard Gaussian designs and independent labels, folding the labels into the design makes the event that a binary w stores every label with robustness radius κ the satisfiability event studied here. Theorem 4 at $\kappa = 0.05$ therefore says that for at least $0.781075N$ random labels, with probability tending to one as $N \to \infty$, no binary weight vector stores all of them with robustness radius 0.05. In floating point the ratio $\alpha_\star/\alpha_{\mathrm{sph}}$ grows from 0.417 at $\kappa = 0$ to 0.556 at $\kappa = 1.5$: against real-valued weights, binary weights give up proportionally less storage as the demanded radius grows.

5 The lower bound

Let $m_N(\kappa)$ be as in Section 1. Krauth and Mézard conjectured the value of the limiting ratio $m_N(\kappa)/N$ for every margin, not only at $\kappa = 0$ [1]; the constant 0.833 is the zero-margin point of the replica-symmetric curve $\alpha_\star(\kappa)$. At $\kappa = 0$ the conjecture is resolved: Ding and Sun [9] proved $\alpha_\star$ is a lower bound with positive probability, the sharp-threshold sequence of Xu [11] and Nakajima and Sun [20] upgrades positive probability to convergence in probability, Huang [10] proved the matching upper bound, and the numerical hypotheses of both proofs are certified [14]. At the five margins of Theorem 4 the upper half holds; elsewhere on the strip it reduces to Condition 4 (Theorem 3(c)). For the lower half, the replica-symmetric free energy is proved at small pattern density by Talagrand [6] and, for a wider class of activations, by Bolthausen, Nakajima, Sun, and Xu [15]; no lower bound matching $\alpha_\star(\kappa)$ is known at $\kappa \neq 0$.

5.1 The second moment at general margin

Write $\mathcal{Z} = \mathcal{Z}_{N,\kappa}$ for the number of $w \in \{-1,+1\}^N$ satisfying the margin condition for all $m = \lceil \alpha N \rceil$ patterns, and $\varpi_\rho(\kappa) = \mathbb{P}(X \geq \kappa,\, Y \geq \kappa)$ for (X,Y) bivariate standard Gaussian with correlation ρ.

Lemma 8. *For every $\kappa \in \mathbb{R}$ and $\alpha > 0$ there is $\delta = \delta(\alpha,\kappa) > 0$ such that*

$$\liminf_{N\to\infty} \frac{1}{N} \log \frac{\mathbb{E}[\mathcal{Z}^2]}{(\mathbb{E}\mathcal{Z})^2} \;\geq\; \delta.$$

Consequently the guarantee of the Paley–Zygmund bound, $\mathbb{P}(\mathcal{Z} > 0) \geq (\mathbb{E}\mathcal{Z})^2/\mathbb{E}[\mathcal{Z}^2]$, is at most $e^{-\delta N + o(N)}$, and no bound depending on $\mathbb{E}\mathcal{Z}$ and $\mathbb{E}[\mathcal{Z}^2]$ alone can certify more, since a distribution placing mass $(\mathbb{E}\mathcal{Z})^2/\mathbb{E}[\mathcal{Z}^2]$ at $\mathbb{E}[\mathcal{Z}^2]/\mathbb{E}\mathcal{Z}$ and the rest at zero matches both moments. For $\kappa \geq 0$ the same conclusion holds for every weighted count $\mathcal{Z}_f = \sum_w \prod_a f(\langle g_a, w\rangle/\sqrt{N})$ with $f \geq 0$ supported on $[\kappa,\infty)$ and $0 < \int f\varphi\, dx < \infty$.

Proof. $\mathbb{E}\mathcal{Z} = 2^N \Psi(\kappa)^m$ by independence over patterns. For a pair w, w' at overlap ρ the fields of each pattern are bivariate Gaussian with correlation ρ, so counting ordered pairs by Hamming distance k,

$$\mathbb{E}[\mathcal{Z}^2] = 2^N \sum_{k=0}^{N} \binom{N}{k} \varpi_{1-2k/N}(\kappa)^m.$$

Fix $\rho_0 \in (0,1)$, let $k^* = \lceil N(1-\rho_0)/2 \rceil$ and $\rho^* = 1 - 2k^*/N$. Keeping the single term k^* and using $\binom{N}{k} \geq e^{Nh(k/N)}/(N+1)$ with h the natural-log binary entropy ($h(p) = h(1-p)$, so the exponent may be written with $(1+\rho_0)/2$),

$$\frac{1}{N} \log \frac{\mathbb{E}[\mathcal{Z}^2]}{(\mathbb{E}\mathcal{Z})^2} \geq \underbrace{h\Big(\tfrac{1+\rho_0}{2}\Big) - \log 2 + \alpha\big[\log \varpi_{\rho_0}(\kappa) - 2\log \Psi(\kappa)\big]}_{=:\,\zeta_{\alpha,\kappa}(\rho_0)} + O(N^{-1}\log N),$$

the error, of order $N^{-1}\log N$, collecting the binomial prefactor, the $|\rho^* - \rho_0| \leq 2/N$ continuity of $\rho \mapsto \log \varpi_\rho$ (differentiable, below), and $|m - \alpha N| \leq 1$. Now $\varpi_0 = \Psi(\kappa)^2$ and $h(1/2) = \log 2$ give $\zeta_{\alpha,\kappa}(0) = 0$. Reflecting the orthant to a joint CDF, $\varpi_\rho(\kappa) = N_2(-\kappa,-\kappa;\rho)$ with N_2 the bivariate normal distribution function, and Plackett's identity [25] $\partial N_2/\partial\rho = \varphi_2$ with the symmetry $\varphi_2(-\kappa,-\kappa;\rho) = \varphi_2(\kappa,\kappa;\rho)$ give $\partial\varpi_\rho/\partial\rho = \varphi_2(\kappa,\kappa;\rho)$ on $(-1,1)$; at $\rho = 0$ this is $\varphi(\kappa)^2$. Since $\frac{d}{d\rho} h(\frac{1+\rho}{2}) = \frac{1}{2}\log\frac{1-\rho}{1+\rho}$ vanishes at 0,

$$\zeta'_{\alpha,\kappa}(0) = \alpha\, \frac{\varphi(\kappa)^2}{\Psi(\kappa)^2} \;>\; 0,$$

so $\zeta_{\alpha,\kappa}(\rho_0) \geq \rho_0\, \zeta'_{\alpha,\kappa}(0)/2 > 0$ for all small $\rho_0 > 0$; fix such a ρ_0 and set $\delta = \zeta_{\alpha,\kappa}(\rho_0)$. For the weighted counts, the diagonal term of $\mathbb{E}[\mathcal{Z}_f^2]$ is $2^N \mathbb{E}[f(X)^2]^m$, so either $\mathbb{E}[\mathcal{Z}_f^2] = \infty$ and the claim is trivial, or $f \in L^2(\varphi\, dx)$ and Mehler's expansion [26] gives $\mathbb{E}[f(X)f(Y)] = \sum_{j\geq 0} \rho^j \langle f, \mathrm{He}_j\rangle^2/j!$, whose ρ-derivative at 0 is $(\int x f\varphi\, dx)^2$; for $f \geq 0$ supported on $[\kappa,\infty)$ with $\kappa \geq 0$ the integrand $xf(x)\varphi(x)$ is nonnegative, and it vanishes almost everywhere only if f vanishes almost everywhere off the origin, which $0 < \int f\varphi\, dx$ excludes; the argument then runs unchanged. □

The mechanism is the asymmetry: pairs of solutions at small positive overlap are exponentially favored over independent pairs at every margin, which is what the AMP-conditioned second moment of [9] is built to circumvent at $\kappa = 0$, and what any lower-bound proof must circumvent at every other margin. At $\kappa = 0$ the failure is classical (see the introduction of [9]); Lemma 8 extends it to every κ and, for $\kappa \geq 0$, to every weighted count. For $\kappa < 0$ the argument gives nothing for some weighted counts: a weight symmetric about the origin, such as the indicator $f = \mathbf{1}_{[-|\kappa|,|\kappa|]}$ of an interval, has $\int x f \varphi \, dx = 0$, so the first-order term vanishes.

5.2 A capacity lower bound at negative margins

Proposition 9. *Let $\kappa < 0$ and let $\alpha_c^{\mathrm{sym}}(r) = \log 2/(-\log \mathbb{P}(|X| \leq r))$ denote the symmetric binary perceptron threshold at width r. For every $\alpha < \alpha_c^{\mathrm{sym}}(|\kappa|)$, with probability $1 - o(1)$ the margin-κ perceptron admits a solution, that is, $m_N(\kappa)/N \geq \alpha$.*

Proof. If $|\langle g_a, w\rangle|/\sqrt{N} \leq -\kappa$ for all a then in particular $\langle g_a, w\rangle/\sqrt{N} \geq \kappa$, so the symmetric solution set at width $|\kappa|$ embeds in the margin-κ solution set. The value $\alpha_c^{\mathrm{sym}}(r)$ is the first-moment threshold of Aubin, Perkins and Zdeborová [16]. Altschuler's Theorem 1.4 [19] states that for every width $r > 0$ and every $\varepsilon > 0$ there is $c > 0$ such that, when the scaled patterns $g_a/\sqrt{N}$ are independent $N(0, N^{-1} I_N)$ vectors and their number is below $\alpha_c^{\mathrm{sym}}(r)N - c \log N$, a binary w with $|\langle g_a, w\rangle|/\sqrt{N} \leq r$ for every a exists with probability at least $1 - \varepsilon$ for N large. Its proof uses the second-moment condition of Abbe, Li and Sly [18, Lemma 3.4] (the second-moment exponent is largest at overlap zero), which they prove for every width in their Section 4.7.1. At fixed $\alpha < \alpha_c^{\mathrm{sym}}(|\kappa|)$ the number of patterns $\lceil \alpha N \rceil$ is eventually below $\alpha_c^{\mathrm{sym}}(|\kappa|)N - c \log N$, so a symmetric solution of width $|\kappa|$ exists with probability $1 - o(1)$. (The lognormal limit of Abbe, Li, and Sly [18] is proved for Rademacher disorder, and Perkins and Xu [17] treat the Gaussian case under a single-critical-point hypothesis, so neither alone covers the model here.) □

Monotonicity gives a second bound. A solution at margin 0 is a solution at every $\kappa < 0$, so $m_N(\kappa) \geq m_N(0)$, and at $\kappa = 0$ the ratio $m_N(0)/N$ converges in probability to $\alpha_\star(0) \approx 0.8330786$ (the lower half by [9] with the sharp-threshold transfer of [20], its numerical hypotheses certified in [14]). Hence for $\kappa < 0$

$$\frac{m_N(\kappa)}{N} \;\geq\; \max\bigl\{\alpha_\star(0),\, \alpha_c^{\mathrm{sym}}(|\kappa|)\bigr\} - o(1) \quad \text{with high probability.}$$

At the two certified negative margins monotonicity is the stronger input: the embedding gives 0.215 at $\kappa = -0.05$ and 0.655 at $\kappa = -0.45$, against the replica-symmetric thresholds 0.889 and 1.558. The embedding takes over only for $\kappa < -0.5757$, where $\alpha_c^{\mathrm{sym}}(|\kappa|)$ passes $\alpha_\star(0)$. Li, Schramm, and Zhou [22] prove lower bounds on the satisfiability threshold of the same one-sided perceptron at negative margin by a different route, with the sharpest statements in the $\kappa \to -\infty$ regime.

5.3 The Ding–Sun condition at general margin

The construction underlying the $\kappa = 0$ lower bound of [9] is written at general margin throughout their Section 2: their equations (34)–(48) define, for parameters $(\kappa_{\mathrm{all}}, \alpha_{\mathrm{all}})$ (their names for the margin and the pattern density), an explicit univariate function $S_{\kappa,\alpha}(\omega)$ on $[\omega_{\min}, 1]$

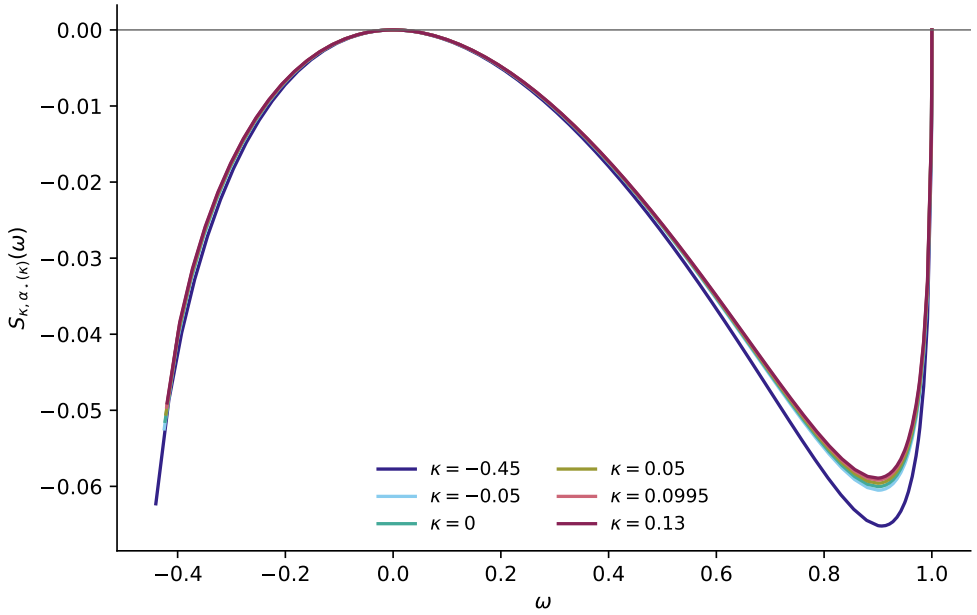


Figure 2: The general-margin Ding–Sun lower-bound condition, evaluated at the five certified margins and $\kappa = 0$: the univariate function $S_{\kappa,\alpha_\star(\kappa)}$ of their equation (48), whose negativity off $\{0, 1\}$ is the numerical hypothesis of their second-moment argument. All six curves are negative off the two roots and nearly coincide.

built from the fixed point $(q_\star, \psi_\star)$ (an entropy term through the two-spin measure $P_{H,D}$, a constraint term through the conditional Mills law φ_ξ, and a convex $\inf_s$ correction), and their Condition 1.2 is the statement $S_{0,\alpha_\star}(\omega) < 0$ for $\omega \notin \{0, 1\}$. The κ-analog of that hypothesis is therefore already defined in their own notation, and its κ-dependent inputs are the fixed point, its uniqueness rectangle and the crossing, which are given by Theorem 3 and the enclosures of Theorem 4.

In floating point, with Gauss–Hermite and conditional Gauss–Legendre quadrature, the function $S_{\kappa,\alpha_\star(\kappa)}$ has the same shape at the five certified margins and at $\kappa = 0$ (Figure 2): at every margin the curve is strictly negative off $\{0, 1\}$, at most $-5 \cdot 10^{-5}$ outside intervals of half-width 0.02 around the two roots (inside them it climbs to its maxima at the roots: at the grid points nearest $\omega = 0$ its maximum lies between $-3.6 \cdot 10^{-6}$ and $-7 \cdot 10^{-7}$, and for $10^{-6} \le 1 - \omega \le 10^{-3}$ it lies between $-0.62\sqrt{1-\omega}$ and $-0.45\sqrt{1-\omega}$; below $1 - \omega = 10^{-6}$ double precision no longer resolves it), the maximum at $\omega = 0$ is degenerate exactly as at $\kappa = 0$, and the six curves have nearly one shape, with $\omega_{\min}$ moving from -0.441 at $\kappa = -0.45$ to -0.421 at $\kappa = 0.13$. The inequality $S_{\kappa,\alpha_\star(\kappa)}(\omega) < 0$ for $\omega \notin \{0, 1\}$ is not proved at any $\kappa \neq 0$.

What separates these floating-point inputs from a general-margin lower bound is the probabilistic body of [9]: the AMP-conditioned moment computation of their Sections 2–7, whose displays carry κ_{all} explicitly; by their Remark 1.5 the assumption $\kappa = 0$ enters only in a few steps, which they indicate. Extending that body to $\kappa \neq 0$ requires, beyond the certified inputs above: a priori bounds at margin (the upper one is free at every κ from the first moment, since

$\mathbb{E}\mathcal{Z} \to 0$ exponentially above the annealed threshold $-\log 2/\log \Psi(\kappa)$; for $\kappa < 0$ the lower one is the monotonicity bound above), the state-evolution inputs (generic in the nonlinearity, hence in κ), and the sharp-threshold transfer of [20] if convergence in probability rather than positive probability is wanted.

Notation

The last column gives the section where the symbol is defined. Symbols used only inside one proof are not listed.

Symbol	Meaning	Section
N	number of binary weights	1
w	weight vector in $\{-1,+1\}^N$	1
g_a	a-th pattern, a standard Gaussian vector in $\mathbb{R}^N$	1
m	number of patterns, $m = \lceil \alpha N \rceil$	1
α	pattern density m/N	1
κ	margin	1
$\varphi,\ \Psi$	standard Gaussian density and upper tail	2
E	inverse Mills ratio φ/Ψ	2
Z	standard Gaussian variable	2
$u(z; q, \kappa)$	$(\kappa - \sqrt{q}\, z)/\sqrt{1-q}$	2
γ	$\sqrt{q/(1-q)}$; at $\kappa = 0$, $u = -\gamma z$ in law	2
$q,\ \psi$	the two order parameters	2
$P,\ R$	the two halves of the fixed-point map, (2)	2
G	replica-symmetric free energy, (3)	2
$G_\star(\alpha)$	G along the fixed-point branch	2
$\alpha_\star(\kappa)$	the zero of $G_\star$; the replica-symmetric threshold	2
$(q_\star, \psi_\star)$	the fixed-point branch as a function of α	2
(q_0, ψ_0)	the fixed point at $\alpha_\star(\kappa)$	2
rectangle clause	the uniqueness and contraction part of Condition 1	2
R_α	the map $q \mapsto R(q, \alpha)$	2
located box	a box certified to contain the fixed point at one α	2
$\ell(\hat{z})$	the local-concavity function of Condition 3	2
$\mu(\hat{z})$	$\mathbb{E}[(\hat{z} + \cosh^2(\sqrt{\psi_0} Z))^{-1}]$	2
A	$1 - q_0$	2
$F(x)$	$E\big((\kappa - x)/\sqrt{A}\big)/\sqrt{A}$	2
$\hat{f}_0,\ d_0$	$-F'/(1 + AF')$ and $\alpha_\star \mathbb{E}[F'(\sqrt{q_0} Z)]$, in the formula for ℓ	2
ℓ_0	the minimum of ℓ	3.1
$\alpha_c(\kappa)$	the bound $2/(\pi \mathbb{E}[(\kappa - Z)_+^2])$ of Nakajima's uniqueness theorem	3.1
$\tau,\ \beta$	$E'(u)$ and $\mu(\hat{z}) - A$ in the formula for ℓ	3.2
located rectangle	on a sub-slab, its $\alpha_\star$ interval times the padded q interval of the located fixed points	3.4
$\mathcal{H}$	$\nabla^2_\lambda \mathcal{S}(\tanh(\lambda_1 \dot{H} + \lambda_2 M), s_0)$ at $\lambda = (1, 0)$	4.4

Symbol	Meaning	Section
η	$E\big((\kappa-\hat{H})/\sqrt{1-q_0}\big)/\sqrt{1-q_0}$	4.1
$\hat{H}$	a Gaussian of variance q_0, independent of $\dot{H}$	4.1
Q	$-\hat{H}/q_0+\big(\kappa-\hat{H}-(1-q_0)\eta\big)/(1-q_0)$	4.2
p_4	$\mathbb{E}\tanh^4(\sqrt{\psi_0}Z)$	4.4
$I_1,\ I_2,\ I_3$	the entries of $\nabla^2\Phi(b_0)$	4.4
$C_1,\ C_2,\ C_3$	second derivatives at $a^\star$ of the constraint term $\mathcal{C}$	4.4
$\dot{H}$	a Gaussian of variance ψ_0	1
M	$\tanh\dot{H}$	1
Λ	a profile, a function of $\dot{H}$ with values in $[-1,1]$	1
$a=(a_1,a_2)$	the moments $(\mathbb{E}[\dot{H}\Lambda],\mathbb{E}[M\Lambda])$	1
K	the moment body, the set of attainable a	4.1
$\mathcal{S}(\Lambda,s)$	Huang's functional of a profile Λ and a tilt $s\geq 0$	4.1
$\mathcal{S}_\star(\lambda_1,\lambda_2)$	$\inf_{s\geq 0}\mathcal{S}(\tanh(\lambda_1\dot{H}+\lambda_2 M),s)$	4.1
$\lambda,\ b$	dual points in $\mathbb{R}^2$	1
$\Phi(b)$	$\mathbb{E}\log 2\cosh(b_1\dot{H}+b_2M)$	1
$H(a)$	entropy term as a function of the moments	4.1
$\mathrm{ent}(\Lambda)$	the entropy term $\mathbb{E}\,h\big(\frac{1+\Lambda}{2}\big)$	4.1
$h(p)$	binary entropy $-p\log p-(1-p)\log(1-p)$	4.1
$D(a)$	$\sqrt{1-a_2^2/q_0}$	4.1
$a^\star$	$(\psi_0(1-q_0),q_0)$, the moment pair of the maximizing profile $\tanh\dot{H}$	1
b_0	the dual point $(1,0)$, at which $\nabla\Phi(b_0)=a^\star$	1
s_0	the optimal tilt $\sqrt{1-q_0}$ at $a^\star$	1
hull	the hull of the boxes located at the two ends of an $\alpha_\star$ interval	2.1
Picard box	a box containing the fixed point for every α of an $\alpha_\star$ interval	4.4
$\mathcal{F}(a,s)$	upper bound for $\mathcal{S}(\Lambda,s)$ at moment $a(\Lambda)=a$, with the maximal entropy $H(a)$	4.1
$\bar{\mathcal{S}}(a)$	$\inf_{s\geq 0}\mathcal{F}(a,s)$	4.1
$v,\ c,\ t$	ray direction, tilt slope and distance along the ray	4.2
$T(v)$	the certified radius in direction v; the star is $\{a^\star+tv:0\leq t\leq T(v)\}$	4.2
$\chi_{v,c}(t)$	$\mathcal{F}(a^\star+tv,\ s_0+ct)$	4.2
Δ	$(\lambda_1-1)\dot{H}+\lambda_2 M$ in the integrand written against b_0	1
band, sector	a radial interval $[t_0,t_1]$ of the rays; one of 768 equal angular ranges of directions	4.2
$V,\ \sigma$	argument of E and $E(V)-V$	Lemma 6
$\bar{B}$	Loewner majorant of $\nabla^2\Phi$ on a Region I band	4.3
$\alpha_{\mathrm{sph}}(\kappa)$	spherical capacity $1/\mathbb{E}[(\kappa-Z)_+^2]$	3.1
$\mathcal{Z}$	number of solutions at margin κ	5
$\varpi_\rho(\kappa)$	$\mathbb{P}(X\geq\kappa,Y\geq\kappa)$, correlation ρ	5
$\zeta_{\alpha,\kappa}(\rho)$	the exponent of the second-moment ratio at overlap ρ	5

Symbol	Meaning	Section
$\delta(\alpha,\kappa)$	a positive lower bound for $\frac{1}{N}\log(\mathbb{E}[\mathcal{Z}^2]/(\mathbb{E}\mathcal{Z})^2)$, Lemma 8	5
$\alpha_c^{\mathrm{sym}}(r)$	symmetric binary perceptron threshold at width r	5
slab	a κ interval of width $5\cdot10^{-4}$ with its rectangle	3
rectangle, box	products of intervals in (α, q), and in (q_0,ψ_0)	3
sub-slab	one of eight κ intervals of width $6.25\cdot10^{-5}$ covering a slab	3
slab rectangle	the rectangle of a slab, its $\alpha_\star$ interval times its q interval	3
contraction product	$\sup_\psi P'\cdot\sup_q\lvert\partial R/\partial q\rvert$ over a rectangle, a bound for $\sup_q(P\circ R_\alpha)'$	3.3
slab's $\alpha_\star$ interval	the enclosure of $\alpha_\star$ on one slab or sub-slab	3
$\alpha_\star$ interval	the enclosure of $\alpha_\star(\kappa)$ of width about $2\cdot10^{-6}$ at one margin (Theorem 4)	3.4
exclusion region	a product of two intervals about $a^\star$ that contains the star	4.2
enclosure of $a^\star$	the image of the hull under $(q_0,\psi_0)\mapsto(\psi_0(1-q_0),q_0)$	4.2
dual enclosure	a certified region of the dual plane containing the optimal dual points	4.2
leaf	a certified sub-cell of a bisected sweep cell	4.2
weak direction	the line through $a^\star$ along an eigenvector of the Hessian of $\bar{\mathcal{S}}$ at $a^\star$, eigenvalue nearest 0	4.2
supplementary arcs	angular ranges with a ray certificate beyond the main radius	4.2
star	the star-shaped neighborhood of $a^\star$ of the ray certificates	4.2
Region I, II, stage 2	the ray certificates on the star; the sweep of K outside the exclusion region; the sweep of the exclusion region minus the star	4.2
star interior	the certificate that the star lies inside K	4.2
$m_N(\kappa)$	largest m for which a solution exists	5
$S_{\kappa,\alpha}(\omega)$	Ding and Sun's univariate function on $[\omega_{\min},1]$	5
ω, $\omega_{\min}$	the variable of $S_{\kappa,\alpha}$ and the lower end of its domain	5

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital.

Declaration of generative AI and AI-assisted technologies

The mathematics, the software, the certificates and the drafting of this paper were done by AI agents, Claude (Anthropic) and Codex (OpenAI), working through the shared memory of the Math Brain project: its code and tests [27], the framework it runs on [29] and its description [28]. The author set the objective, supervised the runs, reviewed the mathematics and accepts responsibility for the paper.

Data availability

No external data were used. The verification programs, certificates and raw records are available at `https://github.com/yspennstate/ising-perceptron-margin/tree/v2.1`, with the certificates in the folder `verification/results`.

References

[1] W. Krauth and M. Mézard, Storage capacity of memory networks with binary couplings, *J. Physique* **50** (1989), 3057–3066.

[2] E. Gardner, The space of interactions in neural network models, *J. Phys. A* **21** (1988), 257–270.

[3] E. Gardner and B. Derrida, Optimal storage properties of neural network models, *J. Phys. A* **21** (1988), 271–284.

[4] J. H. Kim and J. R. Roche, Covering cubes by random half cubes, with applications to binary neural networks, *J. Comput. System Sci.* **56** (1998), 223–252.

[5] M. Talagrand, Intersecting random half cubes, *Random Structures Algorithms* **15** (1999), 436–449.

[6] M. Talagrand, Intersecting random half-spaces: toward the Gardner–Derrida formula, *Ann. Probab.* **28** (2000), 725–758.

[7] R. D. Gordon, Values of Mills' ratio of area to bounding ordinate and of the normal probability integral for large values of the argument, *Ann. Math. Statist.* **12** (1941), 364–366.

[8] M. R. Sampford, Some inequalities on Mill's ratio and related functions, *Ann. Math. Statist.* **24** (1953), 130–132.

[9] J. Ding and N. Sun, Capacity lower bound for the Ising perceptron, *Probab. Theory Related Fields* **193** (2025), 627–715; STOC 2019; arXiv:1809.07742v1, whose numbering is used here.

[10] B. Huang, Capacity threshold for the Ising perceptron, *FOCS 2024*, 1126–1136; arXiv:2404.18902v2, whose numbering is used here.

[11] C. Xu, Sharp threshold for the Ising perceptron model, *Ann. Probab.* **49** (2021), 2399–2415.

[12] D. J. Altschuler and K. Tikhomirov, A note on the capacity of the binary perceptron, arXiv:2401.15092.

[13] S. Nakajima, Uniqueness of RS saddle point for Ising perceptron, arXiv:2512.23195.

[14] Y. Shmalo, The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions, arXiv:2607.11010 (2026); verification repository at `https://github.com/yspennstate/ising-perceptron-capacity/tree/v1.0`.

[15] E. Bolthausen, S. Nakajima, N. Sun and C. Xu, Gardner formula for Ising perceptron models at small densities, *Proceedings of COLT 2022*, PMLR **178** (2022), 1787–1911; arXiv:2111.02855.

[16] B. Aubin, W. Perkins and L. Zdeborová, Storage capacity in symmetric binary perceptrons, *J. Phys. A* **52** (2019), 294003.

[17] W. Perkins and C. Xu, Frozen 1-RSB structure of the symmetric Ising perceptron, STOC 2021; arXiv:2102.05163.

[18] E. Abbe, S. Li and A. Sly, Proof of the contiguity conjecture and lognormal limit for the symmetric perceptron, FOCS 2021; arXiv:2102.13069.

[19] D. J. Altschuler, Critical window of the symmetric perceptron, *Electron. J. Probab.* **28** (2023), 1–28; arXiv:2205.02319.

[20] S. Nakajima and N. Sun, Sharp threshold sequence and universality for Ising perceptron models, *SODA 2023*, 638–674; arXiv:2204.03469.

[21] M. Stojnic, Binary perceptrons capacity via fully lifted random duality theory, arXiv:2312.00073.

[22] S. Li, T. Schramm and K. Zhou, Discrepancy algorithms for the binary perceptron, *STOC 2025*, 1668–1679; arXiv:2408.00796.

[23] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017), 1281–1292.

[24] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[25] R. L. Plackett, A reduction formula for normal multivariate integrals, *Biometrika* **41** (1954), 351–360.

[26] F. G. Mehler, Ueber die Entwicklung einer Function von beliebig vielen Variablen nach Laplaceschen Functionen höherer Ordnung, *J. Reine Angew. Math.* **66** (1866), 161–176.

[27] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, https://github.com/yspennstate/math-brain-public/tree/f96f1dd117d2869e61d01d7090c137548875ce3d.

[28] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, https://github.com/yspennstate/math-brain-public/tree/f96f1dd117d2869e61d01d7090c137548875ce3d/papers/math-brain-2026-10-24.

[29] Y. Shmalo, *Brain Framework*, public software repository, 2026, https://github.com/yspennstate/brain-framework-public/tree/c788e6690a6c40f5637bb1c9f9caef3cf1353a73.